

Week 8 — Strings Inside-Out

Name: ____ Date: ____

Positions start at **0**. Slices stop **before** the end number. And every character is secretly a **number**.

Today's words

Word	What it means
index	a letter's position number — counting starts at 0
len()	counts the characters in a string
slice	a cut-out piece, like <code>word[0:3]</code> — the end index is NOT included
ord() / chr()	<code>ord("A")</code> → 65 reveals the secret number; <code>chr(66)</code> → B goes backwards

Mini ASCII table (you'll need this!)

A	B	C	D	E	F	G	H	I	J	K	L	M
65	66	67	68	69	70	71	72	73	74	75	76	77
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
78	79	80	81	82	83	84	85	86	87	88	89	90

(space = 32)

1 • Address every letter 📌

Write the index under each letter of PYTHON, then answer:

P__ Y__ T__ H__ O__ N__

• `word[2]` = __ • `word[-1]` = _ • `len(word)` = • `word[1:4]` = __

2 • X-ray the alphabet 🔍

Use the table (no computer needed — YOU are the computer):

- `ord("C")` = __ • `chr(72)` = _ • `chr(ord("M") + 1)` = __
- Lowercase letters sit exactly 32 higher. So `ord("c")` = ____

3 • Crack the number message 🕵️

Each number is one character (remember: 32 = space). Decode:

72 • 73 • 32 • 90 • 79 • 69 → ____

🧠 Brain teaser (optional — take it home)

A secret message, in pure numbers. Decode it with your table:

89 79 85 • 32 • 70 79 85 78 68 • 32 • 84 72 69 • 32 • 83 69 67 82 69 84

Then write **your own name** as secret numbers and test it on someone at home. Bring both next week for a shout-out — next week we build a 2,000-year-old cipher on top of exactly this trick.